

BUS33-3

Corporate Finance

The three decisions — invest, finance, return: time value of money, NPV and IRR, the cost of capital and WACC, capital structure, payout policy and value creation

Stop reading companies and start making decisions for them — where to deploy capital, how to fund it, and how much to return.

Albert School — 22 hours

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Preface

In the first two courses of the Finance arc you learned to **read** a company. In **Financial Accounting & Statements** (BUS13-3) you learned the three statements as constructed documents and computed the ratios and working capital that summarise a firm's health. In **Financial Diagnosis & Management Control** (BUS23-3) you turned that literacy into diagnosis: you decomposed profitability with SIG and DuPont, isolated operating performance with **ROIC**, computed the **free cash flow** that valuation actually discounts, and learned to adjust accounts so a comparison is honest. Crucially, you already met the central test of value creation — a firm creates value only when $ROIC > WACC$ — but left WACC as a black box, “the cost of capital, developed in S3”. This is S3.

This course is where you stop describing companies and start **deciding for** them. The whole subject organises around Aswath Damodaran's three corporate-finance decisions:

- **Invest** — where to deploy capital. Which projects earn more than they cost?
- **Finance** — how to fund it. What mix of debt and equity, at what cost?
- **Return** — how much to give back. Dividends, buybacks, or reinvestment?

Every tool here serves one of those three questions. Time value of money, NPV and IRR answer **invest**. The cost of equity (CAPM), the cost of debt and WACC answer **finance** and supply the discount rate that **invest** needs. Capital structure and payout policy complete **finance** and **return**. And the value-creation metrics — ROIC against WACC, and EVA — are the verdict on whether the three decisions, taken together, made the firm worth more.

A word on method, continued from BUS23-3: every idea arrives as a concrete example first and the general principle second, and we keep the same running company, **Lumio**, a mid-size lighting manufacturer, so that every new tool refers back to one coherent set of numbers you already half-know. Lumio's headline figures (€m): revenue 600, EBIT 80, net income 45, effective tax rate 25%, so $NOPAT = 80 \times 0.75 = 60$; net financial debt 220 (financial debt 260, cash 40), book equity 350, invested capital 570, interest expense 20. From BUS23-3 you know its ROIC is $60/570 \approx 10.5\%$ — by the end of this book you will know the WACC that number must beat, and what to do with the surplus it creates.

1 Time value of money

Worked example — why €1,000 next year is not €1,000 today. A supplier offers Lumio a choice: €1,000 now, or €1,000 in one year. The answer is obvious — take it now — but **why**? Because €1,000 today can be placed at, say, 5% and become €1,050 in a year, so €1,000 in a year is worth **less** than €1,000 today. Exactly how much less? Whatever sum, placed at 5% today, grows to €1,000 in a year: $€1,000/1.05 \approx €952$. That €952 is what next year's €1,000 is **worth now**. Every valuation in this book is a more elaborate version of this one division.

A sum of money available today is not equivalent to the same nominal sum available later: money today can be invested to earn a return, so it is worth more. The **time value of money** is the machinery that makes cash flows occurring at different dates comparable — and comparison is the prerequisite for every decision that follows.

Definition 1 (Future and present value) At a periodic rate r , a sum C invested for n periods grows by **compound interest** — interest earning interest — to its **future value**

$$FV = C(1 + r)^n.$$

Conversely, the **present value** of a sum C_n received n periods from now is obtained by **discounting**:

$$PV = \frac{C_n}{(1 + r)^n}.$$

The factor $1/(1 + r)^n$ is the **discount factor**; the rate r used to discount is the **discount rate**.

Compounding and discounting are inverse operations: compounding carries a sum **forward** in time and multiplies, discounting carries it **backward** and divides. The further away a cash flow and the higher the rate, the more discounting shrinks it — which is why distant, risky cash flows contribute so little to value today.

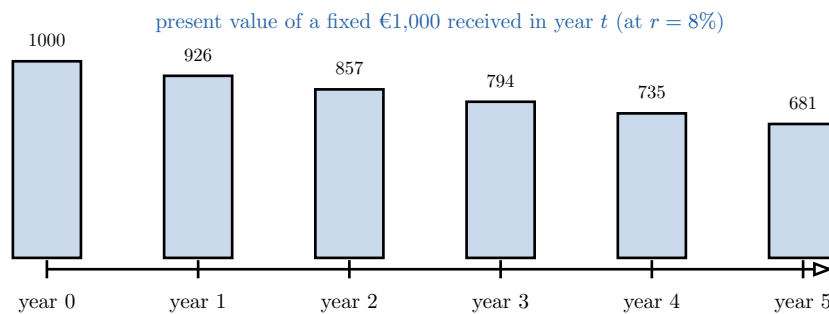


Figure 1: Discounting at work. A flat €1,000 promised in each future year is worth progressively less today: €1,000 now, but only €681 if it arrives in five years. The discount factor $1/(1 + r)^t$ erodes value geometrically with distance and with the rate.

For a **stream** of cash flows $C_1, C_2, \dots, C_n$ occurring at the end of periods 1 through n , present values simply add — each is carried back to today independently and then summed:

$$PV = \sum_{t=1}^n \frac{C_t}{(1 + r)^t}.$$

Three standard streams recur often enough to have closed forms, which save you from summing term by term.

Proposition 2 (Perpetuities and annuities)

- A **perpetuity** paying C at the end of every period forever is worth

$$PV = \frac{C}{r}.$$

- A **growing perpetuity** whose payment grows at rate $g < r$ each period (first payment C) is worth

$$PV = \frac{C}{r - g}.$$

- An **annuity** paying C for n periods is worth

$$PV = \frac{C}{r} \left(1 - \frac{1}{(1 + r)^n} \right).$$

Worked example — the growing perpetuity is the engine of valuation. Suppose Lumio's free cash flow next year is €30 m and is expected to grow at $g = 2\%$ forever, discounted at $r = 8\%$. Its value today is

$$PV = \frac{30}{0.08 - 0.02} = \frac{30}{0.06} = \text{€}500 \text{ m}.$$

This single formula — the **Gordon growth** model — is the terminal-value engine behind almost every company valuation you will build in S4. Notice how sensitive it is: nudge g to 3% and the value jumps to €600 m. Small assumptions about the far future drive large fractions of value, a fragility we return to in the final chapter.

Common pitfall The discount rate must **match** the cash flows in three ways, or the arithmetic is meaningless:

- **Periodicity.** A monthly cash flow is discounted at a monthly rate. An annual rate of 12% is **not** 1% per month compounded — $(1.01)^{12} = 12.7\%$, not 12%.
- **Inflation basis.** Discount **nominal** cash flows (including expected inflation) at a **nominal** rate, and **real** cash flows at a **real** rate. Mixing the two is a classic and large error.
- **Risk.** The rate must reflect the **risk** of these particular cash flows — the subject of Chapters 4–6. Risk-free cash flows are discounted at the risk-free rate; risky ones demand more.

Everything that follows — NPV, the cost of capital, the whole of valuation — is an application of this single principle: a cash flow is worth its discounted value, and values at the same date may be compared and added.

2 Net Present Value and the investment decision

We now make the first of the three decisions: **invest**. An investment project is a set of dated cash flows — an outflow now (and perhaps later) in exchange for inflows in the future. The question is whether the future inflows, **valued today**, exceed what they cost.

Worked example — should Lumio buy the new production line?. Lumio can install an automated line for €100 m today, which will generate net cash inflows of €30 m, €35 m, €40 m and €25 m over the next four years (then be scrapped at nil value). Is it worth it? You **cannot** simply add the inflows ($30 + 35 + 40 + 25 = €130 \text{ m} > €100 \text{ m}$) and declare victory — those euros arrive at different dates and are not comparable. Discount each to today at the firm's 8% cost of capital, then compare with the €100 m outlay.

Definition 3 (Net Present Value) The **Net Present Value** of a project with cash flows $C_0, C_1, \dots, C_n$ (with C_0 usually negative — the initial outlay) discounted at rate r is

$$\text{NPV} = \sum_{t=0}^n \frac{C_t}{(1+r)^t} = C_0 + \frac{C_1}{1+r} + \dots + \frac{C_n}{(1+r)^n}.$$

NPV measures the value created by the project **today**, after fully compensating investors for the time value of money and the risk of the cash flows through the discount rate. A positive NPV is value the project adds to the firm over and above the cost of the capital it ties up.

Worked example — the production line, valued. At $r = 8\%$:

$$\begin{aligned} \text{NPV} &= -100 + \frac{30}{1.08} + \frac{35}{1.08^2} + \frac{40}{1.08^3} + \frac{25}{1.08^4} \\ &= -100 + 27.8 + 30.0 + 31.8 + 18.4 = +€8.0 \text{ m} . \end{aligned}$$

Discounted, the inflows are worth €108 m against a €100 m cost: the line creates about €8 m of value today. Accept it. Note how different this is from the naïve €30 m surplus the undiscounted sum suggested — discounting ate most of it, and at a high enough rate would have turned the project negative.

Proposition 4 (The NPV rule)

- For a **single** project: accept if $\text{NPV} > 0$, reject if $\text{NPV} < 0$.
- Among **mutually exclusive** projects (you can pick only one): choose the one with the **highest** NPV, provided it is positive.

The rule is not arbitrary: a positive-NPV project returns more than the cost of the capital it uses, so taking it raises the value of the firm by exactly the NPV. Maximising NPV **is** maximising shareholder value — the objective the rest of the book serves.

$$\Sigma \text{ discounted inflows} = 108 > 100 \text{ outlay} \Rightarrow \text{NPV} = +8$$

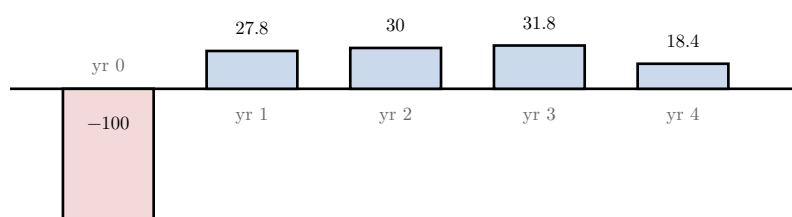


Figure 2: The production line as discounted cash flows (€m). The €100 m outlay at year 0 is set against the **present values** of the four inflows — not their face values. Their sum, €108 m, exceeds the outlay by the €8 m NPV.

The discount rate used is the **cost of capital** (Chapters 4–6), reflecting the risk of the project’s cash flows — for a project of average firm risk, the firm’s WACC. A riskier project demands a higher rate, and the same cash flows then create less (or negative) value.

Common pitfall Build NPV on **incremental, after-tax cash flows**, and watch four traps:

- **Sunk costs** are irrelevant — money already spent (a feasibility study) does not change with the decision and must be excluded.
- **Opportunity costs** are relevant — using land Lumio already owns has the cost of what that land could otherwise earn.
- **Cash, not accounting profit** — add back depreciation, subtract capex and working-capital investment, exactly as in the free-cash-flow build of BUS23-3.
- **Financing flows belong in the discount rate, not the cash flows** — do not subtract interest from the project’s cash flows **and** discount at the WACC; that double-counts the cost of debt.

3 Internal Rate of Return

NPV answers “how much value?” Managers often also want a single **rate of return** they can compare against the cost of capital and quote without restating a discount rate. That number is the IRR.

Definition 5 (Internal Rate of Return) The **Internal Rate of Return** of a project is the discount rate that makes its NPV exactly zero:

$$\sum_{t=0}^n \frac{C_t}{(1 + \text{IRR})^t} = 0.$$

It is the project’s own break-even rate — the return it earns on the capital it ties up, expressed as a percentage.

Worked example — the production line's IRR. For Lumio’s line, the rate that zeroes the NPV is about $\text{IRR} \approx 10.8\%$. Read it against the 8% cost of capital: the project earns 10.8% on its capital, comfortably above the 8% that capital costs, so it creates value — the same verdict NPV gave. The two agree because, for this project, NPV is positive precisely while the discount rate sits below 10.8%.

Proposition 6 (When IRR and NPV agree) For a **conventional** project — one initial outflow followed only by inflows, so the cash flows change sign exactly once — the IRR rule “accept if $\text{IRR} > r$ ” gives the **same accept/reject verdict** as the NPV rule. This is because such a project’s NPV falls monotonically in the discount rate and crosses zero exactly once, at the IRR.

The agreement is only on the **yes/no** decision for a single conventional project. For **ranking** projects, and for unconventional cash flows, IRR can mislead in three distinct ways.

Worked example — scale: the higher IRR, the lower value. Lumio weighs two mutually exclusive projects at $r = 8\%$:

Project	Outlay	Inflow yr 1	IRR	NPV at 8%
Small	€10 m	€13 m	30%	€2.0 m
Large	€100 m	€118 m	18%	€9.3 m

IRR prefers Small ($30\% > 18\%$); NPV prefers Large ($€9.3 \text{ m} > €2.0 \text{ m}$). IRR is a **rate** and is blind to **size** — 30% of a small base is less money than 18% of a large one. If Lumio can take only one, it should take Large: it adds more value. NPV ranks correctly; IRR does not.

The three ways the IRR rule breaks:

- **Scale.** As above, IRR ignores project size and can rank a small project above a larger, more valuable one.
- **Timing.** Projects whose cash flows arrive on different schedules can rank differently under IRR than under NPV, and the IRR ranking can even **reverse** as the cost of capital changes — they swap at a **crossover rate** where their NPV profiles intersect.
- **Multiple or no IRR.** A project whose cash flows change sign **more than once** (an outflow, inflows, then a large clean-up outflow) can have several IRRs, or none — leaving “accept if $\text{IRR} > r$ ” undefined.

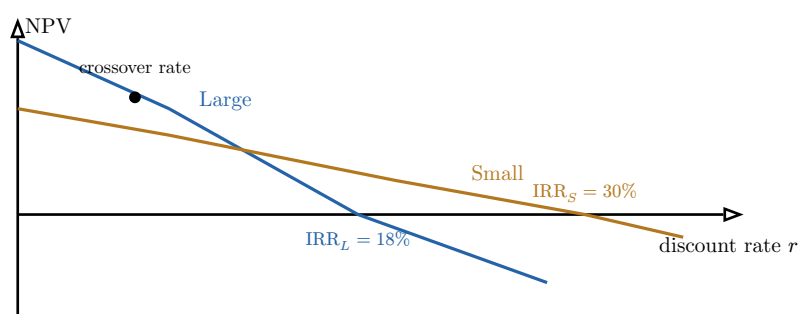


Figure 3: NPV profiles of two mutually exclusive projects. Each curve crosses zero at its own IRR (Small has the higher IRR). But to the **left** of the crossover rate, Large has the higher NPV — so at a low cost of capital the IRR ranking is wrong. NPV reads off the correct project at the firm’s actual r .

Common pitfall Two persistent errors with IRR:

1. **Treating the higher IRR as the better project.** For mutually exclusive choices, rank by NPV, not IRR — the scale and timing effects above make the IRR ranking unreliable.
2. **Implicit reinvestment.** The IRR computation implicitly assumes interim cash flows are reinvested **at the IRR itself**, which is unrealistic for a high-IRR project. NPV assumes reinvestment at the cost of capital — the economically correct rate.

NPV is the **superior** rule: it measures value added directly, in currency, at the relevant cost of capital, and never multiplies or vanishes. IRR survives as a **communication** device — a return figure that can be compared against the cost of capital across projects without restating the discount rate — and as an honest accept/reject test for a single conventional project. Know its failure modes and default to NPV whenever they bite.

4 Cost of equity via CAPM

The discount rate that NPV and IRR both needed is the **cost of capital** — and to build it we must first price each source of finance separately. We begin with equity. Shareholders are

residual claimants: they are paid last and bear the firm's risk, so they require a return for it. But **which** risk?

Worked example — the risk that gets paid for. A shareholder who holds only Lumio bears all of Lumio's ups and downs. But a shareholder holding a **diversified portfolio** of hundreds of stocks barely feels a Lumio-specific shock — a factory fire, a product recall — because other holdings offset it. That **firm-specific** (diversifiable) risk can be diversified away for free, so the market does not reward bearing it. What **cannot** be diversified away is the risk Lumio shares with the whole market — recessions, rate shocks. **Only this systematic risk is priced.** This is the insight the CAPM formalises.

Definition 7 (CAPM — the cost of equity) The Capital Asset Pricing Model gives the required return on a firm's equity as

$$r_e = r_f + \beta(r_m - r_f),$$

where

- r_f is the **risk-free rate**;
- β (beta) is the stock's sensitivity to market movements — the systematic risk;
- $r_m - r_f$ is the **equity risk premium** (ERP): the excess return investors demand for holding the market portfolio over the risk-free asset.

The product $\beta(r_m - r_f)$ is the stock's own **risk premium**.

Beta is the bridge to your Math S2 toolkit: it is the covariance of the stock's return with the market's, divided by the variance of the market's return, $\beta = \text{Cov}(r_i, r_m) / \text{Var}(r_m)$ — a standardised measure of how strongly the stock moves with the market. $\beta = 1$ means it moves one-for-one with the market; $\beta > 1$ amplifies market moves (cyclical firms); $\beta < 1$ dampens them (defensive firms such as utilities).

Worked example — Lumio's cost of equity. Take $r_f = 3\%$ (a long government bond yield), $\beta = 1.2$ (lighting is mildly cyclical), and an equity risk premium of $r_m - r_f = 6\%$. Then

$$r_e = 3\% + 1.2 \times 6\% = 3\% + 7.2\% = 10.2\%.$$

Lumio's shareholders require about 10.2% a year. Notice this already exceeds its 10.5% ROIC by very little on the equity side alone — a first hint that Lumio's value creation is real but not lavish.

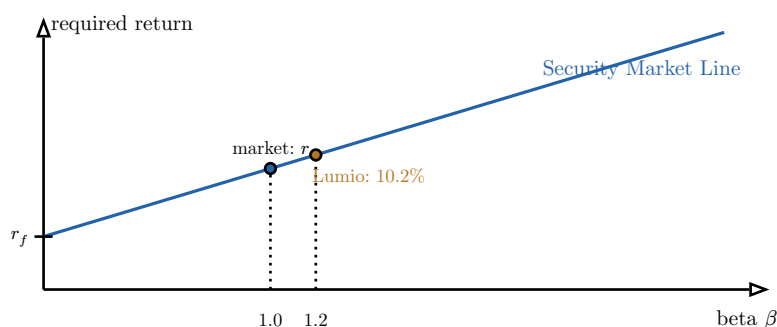


Figure 4: The Security Market Line. Required return rises linearly with systematic risk β : it starts at the risk-free rate ($\beta = 0$), passes through the market return at $\beta = 1$, and places Lumio slightly above the market because its $\beta = 1.2$.

Each input demands judgment — CAPM is a disciplined framework, not a number that falls out of a database:

- **Risk-free rate.** Proxied by a government bond yield, with maturity matched to the horizon of the cash flows being discounted (a long bond for long-lived projects).
- **Beta.** Estimated by regressing the stock's returns on market returns (your Math S2 regression), or sourced from a data provider. It shifts with the estimation window, the return frequency (daily vs monthly), and the index chosen, and is often **adjusted toward 1** because betas tend to revert. For an unlisted firm or division, take a **comparable** firm's beta and re-lever it to the target structure.
- **Equity risk premium.** Estimated either from long-run **historical** average excess returns or **implied** from current market prices; the two can differ by several points, so the choice must be argued, not assumed.

Common pitfall CAPM prices **systematic** risk only. Do not inflate r_e for risks that diversification removes — a single-product firm's product risk, say — those belong in the **cash-flow** forecast (lower or probability-weighted flows), not in the discount rate. And never use a **single** firm-wide r_e to discount a project of very different risk: a stable utility evaluating a speculative tech venture must use the venture's higher beta, or it will accept value-destroying projects that a blended rate wrongly waves through.

5 Cost of debt

The second source of finance is debt — and it is cheaper than equity for two reasons: lenders are paid **before** shareholders and bear less risk, and interest is **tax-deductible**. Both effects must be quantified.

Worked example — what Lumio's debt really costs. Lumio's bonds trade at a yield of 5%: that is the return lenders require today, the **pre-tax** cost of debt — not the historical coupon printed on old bonds, which is irrelevant to a decision made now. But 5% overstates the true cost. Because the €1 of interest Lumio pays is deductible, at a 25% tax rate it cuts the tax bill by €0.25. The net cost of that interest is only €0.75 — an **after-tax** rate of $5\% \times (1 - 0.25) = 3.75\%$. Debt is cheap, and the tax system makes it cheaper.

Definition 8 (Pre-tax cost of debt) The **pre-tax cost of debt** r_d is the return a lender requires **today**. Estimate it as either:

- the **yield to maturity** (YTM) on the firm's traded bonds — the single discount rate equating the bonds' price to the present value of their promised coupons and principal; or
- the **risk-free rate plus a credit spread**, $r_d = r_f + \text{spread}$, where the spread compensates for default risk and widens as credit quality falls.

The YTM is itself a time-value-of-money construction: it is the bond's IRR. The credit spread is set by the firm's default risk, which you can read off the solvency and coverage ratios from BUS13-3 and BUS23-3 — a firm with interest coverage of $1.5 \times$ borrows at a far wider spread than one covering $8 \times$.

Definition 9 (After-tax cost of debt and the tax shield) Because interest is tax-deductible, the cost of debt relevant to the firm is the **after-tax cost**:

$$r_d(1 - \tau),$$

where τ is the marginal corporate tax rate. The tax saved because interest is deductible is the **debt tax shield**: on interest expense I , the saving is

$$\tau \times I \text{ per period.}$$

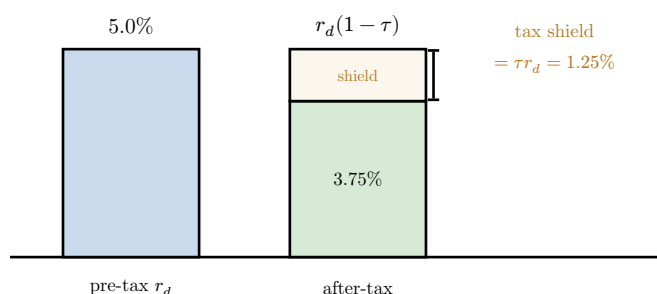


Figure 5: The tax shield on Lumio’s debt. A 5% pre-tax cost becomes a 3.75% after-tax cost because the deduction returns $\tau r_d = 0.25 \times 5\% = 1.25\%$ to the firm. The shield is the wedge between the two bars and the mechanism by which leverage can add value (Chapter 7).

The tax shield matters twice over. As a **cost**, it makes debt cheaper than its stated rate — the after-tax rate is what enters WACC next chapter. As a **source of value**, it is the channel through which capital structure can change firm value at all, the hinge of the Modigliani–Miller analysis in Chapter 7.

Common pitfall Use the **current, forward-looking, after-tax** cost of debt for decisions — the rate the firm would pay to borrow **today**. Do not use the average coupon on legacy debt issued years ago at different rates; it is a historical accident, not the cost of new capital. And the shield is only worth τr_d **if the firm is actually paying tax**: a loss-making firm with no taxable profit gets no shield this year, so its after-tax cost of debt is simply its pre-tax cost.

6 Weighted Average Cost of Capital

A firm financed by both equity and debt must earn enough to satisfy **both** claimant groups. The WACC blends the two costs into the single hurdle rate that the invest decision has been waiting for.

Worked example — Lumio's blended hurdle rate. Lumio is financed by equity worth $E = \text{€}450$ m (market value) and debt worth $D = \text{€}260$ m, so $V = 710$ m. Equity costs $r_e = 10.2\%$, debt costs $r_d = 5\%$ pre-tax (3.75% after the 25% tax). The firm’s cost of capital is the **weighted average** of the two, by their share of total financing:

$$\text{WACC} = \frac{450}{710} \times 10.2\% + \frac{260}{710} \times 3.75\% = 6.47\% + 1.37\% \approx 7.8\%.$$

This is the 8%-ish number BUS23-3 assumed. Lumio’s ROIC of 10.5% clears it — the firm creates value — but by a spread of under three points, which the rest of the book will learn to defend, widen, or doubt.

Definition 10 (WACC) With market values of equity E and debt D , total $V = E + D$, the **Weighted Average Cost of Capital** is

$$\text{WACC} = \frac{E}{V}r_e + \frac{D}{V}r_d(1 - \tau),$$

where r_e is the cost of equity, r_d the pre-tax cost of debt, and τ the corporate tax rate. It is the minimum return the firm's assets must earn to satisfy all capital providers — and the discount rate for a project of the same risk as the firm as a whole.

The weights are **market-value** (or target) weights, never book weights, because they represent the cost of raising capital **today**: book equity is a historical accounting residual, while the cost of equity is a forward-looking required return that applies to equity's market value.

How does **leverage** move WACC? Replacing expensive equity with cheaper after-tax debt lowers the average — at first. But more debt raises the riskiness of **both** claims: equity becomes more volatile (its beta rises with leverage), and at high debt levels lenders demand wider spreads too. So r_e and r_d both climb as $\frac{D}{V}$ rises, eventually offsetting and then overwhelming the benefit of debt's lower rate. The result is a **U-shaped WACC curve**.

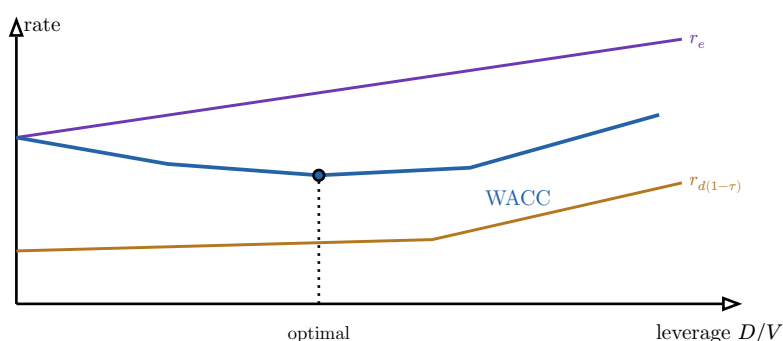


Figure 6: WACC against leverage. As debt replaces equity, the cheap after-tax cost of debt pulls the average down, but the rising cost of **both** equity and debt pulls it back up. WACC is minimised at an intermediate leverage — the same optimum the capital-structure trade-off identifies in the next chapter.

Common pitfall Three WACC errors that quietly corrupt every downstream valuation:

- **Book weights.** Using book equity instead of market equity — especially for a firm whose market value far exceeds book — badly misweights the average.
- **Wrong-risk discounting.** WACC is the right rate only for projects of **firm-average risk**. Discounting a safe cost-saving project and a risky new-market launch at the same WACC over-rewards the risky one and starves the safe one.
- **Double-counting the shield.** The tax benefit of debt is **already** in WACC via $r_d(1-\tau)$. Do not **also** add interest tax shields back into the project's cash flows — pick one method (WACC **or** adjusted present value), never both.

7 Capital structure

WACC's U-shape already told us that the financing mix can matter. Capital structure theory explains **why**, and starts — paradoxically — by proving that in an idealised world it does **not** matter at all. That benchmark is what makes the real-world frictions legible.

Proposition 11 (Modigliani–Miller irrelevance) In a **frictionless** market — no taxes, no bankruptcy costs, no information asymmetry, no transaction costs — the total value of a firm is **independent** of how it is financed. Slicing the same operating cash flows into debt and equity claims does not change the size of the pie:

$$V_{\text{levered}} = V_{\text{unlevered}}.$$

Worked example — why the pie does not grow by slicing it. Imagine Lumio’s operations throw off a fixed stream of cash. Financing it 100% with equity, the shareholders receive the whole stream. Financing it half with debt, lenders take a fixed slice and shareholders take the volatile remainder — but the **two slices still sum to the same stream**. In a world with no taxes and no bankruptcy, an investor could even replicate any leverage themselves by borrowing in their own account (“homemade leverage”), so they will not pay a premium for the firm to do it. Value comes from the **assets**, not the financing. MM is a benchmark, not a description of reality — its job is to tell us that **if** capital structure matters, it matters only through the frictions it assumes away.

Relaxing the assumptions one at a time reveals the forces that make structure matter:

- **Taxes (push toward more debt).** Interest deductibility creates the tax shield of Chapter 5. Each euro of permanent debt adds roughly $\tau \times$ that debt in present-value shield, raising firm value as leverage rises.
- **Distress costs (push toward less debt).** Higher leverage raises the probability of financial distress and bankruptcy, which carry **direct** costs (legal and administrative fees) and, larger, **indirect** costs (lost customers and key staff, fire-sale asset disposals, managers firefighting instead of investing).
- **Agency costs (cut both ways).** Debt can **discipline** managers by committing free cash flow to interest, curbing empire-building (a benefit). But heavy debt also breeds **conflicts** between shareholders and creditors — risk-shifting, under-investment — that destroy value (a cost).

Definition 12 (The trade-off theory) Firm value as a function of leverage is the unlevered value **plus** the present value of the tax shield **minus** the present value of expected distress and agency costs:

$$V_{\text{levered}} = V_{\text{unlevered}} + \text{PV}(\text{tax shield}) - \text{PV}(\text{distress costs}).$$

The shield grows roughly linearly with debt; distress costs accelerate as debt rises. Their net defines an interior **optimal capital structure** — the leverage that maximises value (equivalently, minimises WACC).

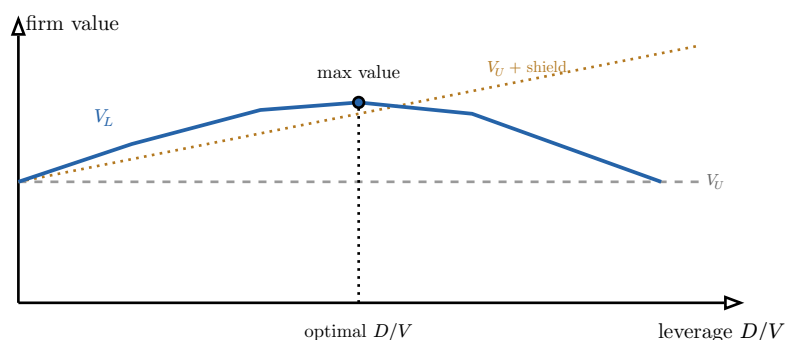


Figure 7: The trade-off theory. The tax shield (dotted) would raise value without limit, but expected distress costs bend the actual value curve (solid) back down. Value peaks at an interior optimum — too little debt forgoes the shield, too much invites distress.

Definition 13 (Over- and under-leverage) Relative to its optimal range, a firm is **over-leveraged** when distress and agency costs dominate — thin interest coverage, constrained investment, fragile to shocks — and **under-leveraged** when it leaves an unused tax shield on the table and forgoes debt’s disciplining effect, often hoarding cash. Diagnosing which means locating the firm against its target using the **leverage and coverage ratios** of BUS13-3 and BUS23-3 (debt-to-equity, net-debt-to-EBITDA, interest coverage) and its business risk.

Worked example — is Lumio over- or under-leveraged? Lumio’s interest coverage is $80/20 = 4.0 \times$ and its net-debt-to-EBITDA is about $220/120 \approx 1.8 \times$ — comfortable for a stable manufacturer. It is, if anything, **mildly under-leveraged**: a manufacturer with steady cash flows could likely carry more debt, capture more tax shield, and lower its WACC below 7.8% without serious distress risk. A volatile tech firm with the **same** ratios, however, might be at its limit — the safe debt level depends on **business risk**, not the ratios alone.

Common pitfall The optimum is a **range**, not a knife-edge, and it is **industry- and risk-specific**. A stable utility safely carries leverage that would bankrupt a cyclical, asset-light software firm. Never benchmark a company’s leverage against the whole market — only against peers of similar business risk and asset tangibility (asset-heavy firms borrow more because their assets back the debt). And remember the shield is worthless to a firm that pays no tax: do not prescribe debt to a chronic loss-maker.

8 Payout policy

The third decision — **return** — arises once a firm generates cash beyond its profitable investment needs. How much should go back to shareholders, and through which mechanism? The prior question is always **retention vs distribution**.

Definition 14 (The retention-vs-distribution rule) Cash should be **retained and reinvested** only in projects that earn more than the cost of capital — i.e. positive-NPV projects, equivalently those with $ROIC > WACC$. Cash with no such home should be **distributed**: retaining it to earn the WACC (or less) inside the firm creates no value the shareholder could not create alone. Payout policy is what to do with the residual after all positive-NPV investment is funded.

This connects the three decisions into one logic: **invest** claims every positive-NPV euro; **finance** sets the WACC that defines “positive”; **return** distributes the rest. A firm distributing cash while passing up positive-NPV projects, or hoarding cash with no projects, has its policy backwards.

Definition 15 (Payout mechanisms)

- A **dividend** is a cash distribution per share, typically paid on a regular schedule.
- A **share buyback** (repurchase) returns cash by the firm buying its own shares in the market, reducing the share count — so each remaining share owns a larger slice of the firm.

In a frictionless MM world the **choice** is irrelevant: a shareholder wanting cash can sell shares to manufacture a “homemade dividend,” and a euro paid out is a euro the share price drops. The choice matters because of two frictions:

- **Taxes.** Dividends and the capital gains realised through buybacks are often taxed **differently**, and at different times — buybacks defer the tax until the investor sells, dividends tax on receipt. The tax-favoured mechanism depends on jurisdiction and shareholder type.
- **Signalling.** Because cutting an established dividend is painful and punished by the market, **raising** the regular dividend is a **credible signal** of confidence in durable future cash flows — managers will not commit unless they believe it. Buybacks are more **flexible** and **discretionary**, so they signal less commitment but suit lumpy or uncertain surplus cash; they also avoid locking the firm into a payout it may later have to cut.

Worked example — two firms, two right answers. A mature, cash-rich Lumio with stable cash flows and few high-return projects suits a **steady, growing dividend**: it can credibly commit, and its income-oriented shareholders value the regularity. A young, volatile growth firm with lumpy surplus cash and a tax-sensitive, capital-gains-oriented shareholder base suits **buybacks**: flexible, tax-deferring, and not a promise it might have to break. Same surplus, opposite mechanisms — driven by growth stage, cash-flow stability, and the shareholder base.

Common pitfall Buybacks do **not** mechanically “create value” by raising earnings per share. EPS rises only because the share count falls; if the shares are repurchased **above** their intrinsic value, the buyback **destroys** value for continuing shareholders even as EPS ticks up. Judge a buyback like any investment — by price versus value, not by its effect on EPS. And a dividend funded by **new borrowing** rather than free cash flow (recall Penumbra in BUS23-3) is a warning sign, not a sign of health.

9 Value-creation metrics

We can now close the loop opened in BUS23-3. A firm creates value only when it earns more on its invested capital than that capital costs — and we finally have both halves of the comparison: ROIC (from BUS23-3) and WACC (Chapter 6).

Definition 16 (The ROIC-vs-WACC test) With $ROIC = NOPAT / \text{invested capital}$ (BUS23-3) and WACC the cost of capital (Chapter 6), the firm

- **creates** value when $ROIC > WACC$,
- is **value-neutral** when $ROIC = WACC$,
- **destroys** value when $ROIC < WACC$.

The gap $\text{ROIC} - \text{WACC}$ is the **economic spread**.

Definition 17 (Economic Value Added) Economic Value Added (EVA) expresses the same test as a **currency** amount per period:

$$\text{EVA} = (\text{ROIC} - \text{WACC}) \times \text{invested capital} = \text{NOPAT} - \text{WACC} \times \text{invested capital}.$$

The term $\text{WACC} \times \text{invested capital}$ is the **capital charge** — the euro cost of all the capital tied up. Positive EVA is value created over the period; negative EVA is value destroyed, however healthy the accounting profit looks.

Worked example — Lumio's verdict. Lumio: $\text{ROIC} = 10.5\%$, $\text{WACC} \approx 7.8\%$, invested capital €570 m. The spread is $10.5\% - 7.8\% = 2.7\%$, so

$$\text{EVA} = 2.7\% \times 570 \approx \text{€}15.4 \text{ m per year.}$$

Equivalently, $\text{NOPAT} - \text{WACC} \times \text{IC} = 60 - 0.078 \times 570 = 60 - 44.5 = \text{€}15.5 \text{ m}$ (rounding). Lumio earns €60 m of after-tax operating profit but is **charged** €44.5 m for the capital that produced it; the €15 m residual is genuine economic value. A firm reporting the same €45 m net income but with $\text{ROIC} < \text{WACC}$ would be **destroying** value despite its positive bottom line — the distinction accounting profit cannot make.

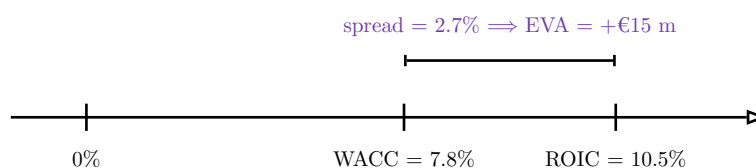


Figure 8: Value creation is a **comparison**, not a level. Lumio's 10.5% ROIC clears its 7.8% WACC, so each euro of the €570 m invested capital earns a 2.7% surplus — €15 m of EVA a year. The same ROIC against a 12% WACC would destroy value despite looking healthy in isolation.

EVA charges the firm for **all** capital employed — including equity, which accounting profit treats as free — at its true cost. This is precisely why a company can report positive net income yet negative EVA: net income subtracts the cost of debt (interest) but never the cost of equity. Computing ROIC and EVA turns the cost of capital from Chapters 4–6 into a verdict on whether the company was **worth operating**.

Common pitfall EVA inherits every measurement choice in ROIC and WACC. Use **NOPAT** (operating profit after tax, before interest), not net income, and keep non-operating cash out of invested capital (the BUS23-3 warnings). On the WACC side, a small error in beta or the ERP can swing the capital charge by millions and flip the EVA sign — so a borderline EVA verdict should be reported as a **range**, not a false-precision point estimate.

10 The integrated investment case

Everything converges here. The three decisions — **invest**, **finance**, **return** — are not taken in isolation but **together**, for one company, using the accounting-adjusted inputs you built in BUS23-3 (normalised earnings, free cash flow, ROIC). This is the deliverable the whole Finance

arc has been building toward, and the input to Valuation (S4) and Corporate Finance Leadership (S6).

Definition 18 (The integrated case) For a single company, assemble the three decisions on **one consistent set of assumptions**:

1. **Invest.** Build the project or strategy as dated, incremental, after-tax **free cash flows** (BUS23-3) and evaluate it with **NPV** at the appropriate discount rate, cross-checked with **IRR** (Chapters 2–3).
2. **Finance.** Choose a financing mix; compute its **WACC** (Chapter 6); assess how the proposed leverage moves the cost of capital and where it sits against the firm's **optimal structure** (Chapter 7). Feed this WACC back as the discount rate in step 1.
3. **Return.** Justify a **payout policy** for the cash not absorbed by positive-NPV investment, consistent with the firm's growth, cash-flow stability, and shareholder base (Chapter 8).

The case is coherent only if the **same** inputs, risk assumptions, and tax rate run through all three — the WACC that discounts the project must be the WACC implied by the financing mix, and the cash distributed must be what is left after funding every positive-NPV project.

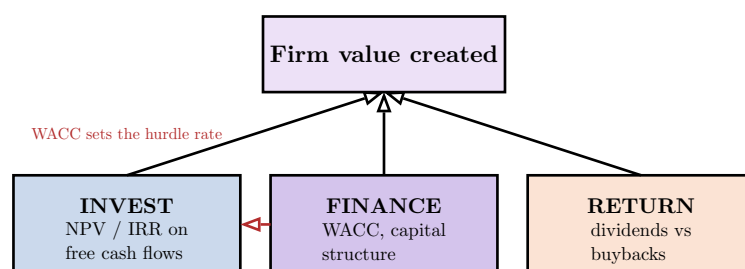


Figure 9: The three corporate-finance decisions feed one objective — value created — and interlock. Finance supplies the WACC that Invest discounts at (red arrow); Invest claims every positive-NPV euro; Return distributes only the residual. The case is sound only when one consistent set of assumptions flows through all three.

Worked example — Lumio's integrated case, in one paragraph. **Invest:** the €100 m production line has NPV +€8 m at the 7.8% WACC and an IRR of 10.8% — accept. **Finance:** Lumio's current WACC is 7.8% at $D/V \approx 37\%$; it is mildly under-leveraged, so funding part of the line with new debt would capture more tax shield and nudge WACC down — strengthening the very NPV that justified the project. **Return:** after funding the line and its other positive-NPV projects, Lumio still generates surplus cash; given stable flows and an income-oriented shareholder base, a steady, modestly growing dividend fits better than buybacks. The three decisions reinforce each other: cheaper financing improves the investment case, and only the genuine residual is returned.

10.1 Critiquing a capital allocation proposal

A finished case is not a verdict to be trusted but an argument to be **stress-tested**. The most valuable skill in corporate finance is knowing **where a model is weakest** — because a positive NPV is only as good as the assumptions underneath it.

Definition 19 (A disciplined critique) Interrogate any capital-allocation proposal along three axes:

1. **Which assumptions drive the NPV?** Identify the few inputs the conclusion is most sensitive to — typically the **growth rate**, **margins**, **terminal value**, and **discount rate** — by varying each and watching the NPV move (sensitivity and scenario analysis). A project whose positive NPV survives only at the optimistic end of one assumption is fragile.
2. **Where is the cost of capital weakest?** The WACC rests on judgment at every input — **beta**, the **equity risk premium**, the **target weights**, the **credit spread** (Chapters 4–6). A small change in any can flip the accept/reject decision, so the estimate’s soft spots must be named.
3. **What can the model not capture?** Discounted cash flow ignores **real options** (the value of waiting, expanding, or abandoning), **strategic interactions** (competitor responses), and risks not expressible as a discount-rate bump. These live outside the spreadsheet and must be argued qualitatively.

Worked example — stress-testing Lumio's production line. The line’s +€8 m NPV is thin. Re-run it: at a 9% WACC instead of 7.8% it turns **negative** — so the conclusion hangs on the cost-of-capital estimate, exactly the input resting on the most judgment (beta, ERP). The year-4 scrap assumption and the €40 m year-3 inflow also swing it. The honest recommendation is therefore **not** “NPV is €8 m, accept,” but “the line creates value **if** the cost of capital is at or below 8.5% and the volume ramp holds — and here is what each assumption is worth.”

Common pitfall A recommendation is defended **not by its point estimate** but by knowing which inputs it depends on and how far it survives their plausible range. Beware the false precision of an NPV quoted to the nearest euro on top of a 6% guessed equity risk premium and a 2% terminal growth rate pulled from the air — the apparent rigour of the arithmetic hides the fragility of the assumptions. And beware **advocacy modelling**: a sponsor who wants the project will find a growth rate that makes NPV positive. The critic’s job is to surface that assumption and price it.

The thread of this book. BUS13-3 taught you to read a company and BUS23-3 to diagnose it; this course taught you to **decide for** it, through Damodaran’s three decisions. **Time value of money** makes cash flows at different dates comparable; **NPV** turns that into the invest rule, with **IRR** as its communicable but fallible cousin. To discount, we priced capital: the **cost of equity** via CAPM (systematic risk only), the after-tax **cost of debt** with its tax shield, and their blend, **WACC** — the hurdle rate BUS23-3 had assumed. **Capital structure** explained when the financing mix changes value (the tax-shield-versus-distress trade-off), and **payout policy** when and how to return the residual cash. The **ROIC-vs-WACC** test and **EVA** delivered the verdict on value creation, and the **integrated case** wove invest, finance and return into one coherent argument — which a disciplined **critique** then stress-tests. This invest–finance–return framework, discounted at a defensible cost of capital, is exactly what **Valuation** (S4) and **Corporate Finance Leadership** (S6) will assume you can build and defend.